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SIT225: Data Capture Technologies

Activity 6.1: Plotly data dashboard

Plotly Dash apps give a point-&-click interface to models written in Python, vastly expanding the notion of what's possible in a traditional "dashboard". With Dash apps, data scientists and engineers put complex Python analytics in the hands of business decision-makers and operators. In this activity, you will learn basic building blocks of Plotly to create Dash apps.

Hardware Required

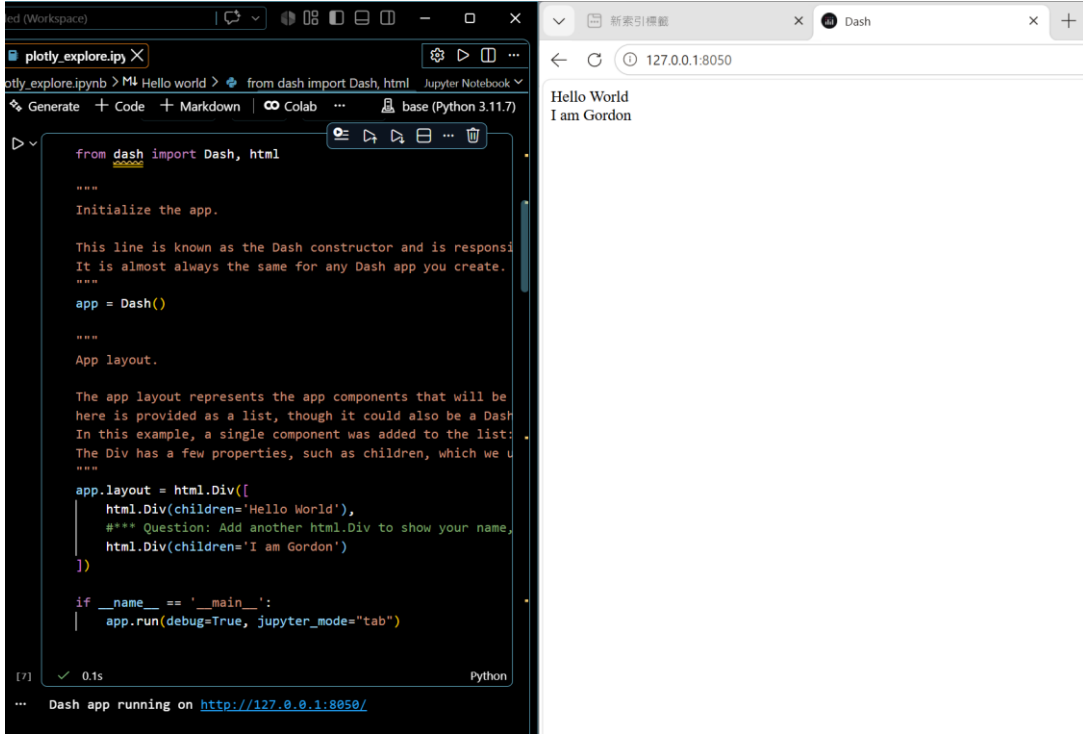
No hardware is required.

Software Required

Plotly library and Dash module
Python 3

Steps

Step	Action
1	Install Plotly and dash using the command below in the command line. <pre>\$ pip install plotly dash</pre> You can download Jupyter Notebook from here (https://github.com/deakin-deep-dreamer/sit225/blob/main/week_6/plotly_explore.ipynb) and run all the cells. The Notebook contains multiple sections such as Hello World which follows a sample code in a following cell. If you run the Hello world cell it will show Plotly Dash web page. The cell also includes a Question (**** Question) which you will need to carry out to get a modified output. You will need to capture the output and share the screenshot in the following steps.

2	Question: Hello World cell has a question - add another html.Div to show your name, and re-run the cell for output. You will need to update the code, run the cell, capture the screenshot of the output and paste it here. Answer:  The screenshot shows a Jupyter Notebook on the left and a web browser on the right. The notebook cell contains Python code for a Dash application. The code imports Dash and html, initializes the app, and sets the layout to a Div containing two other Divs: one with 'Hello World' and another with 'I am Gordon'. The code also includes a comment: '*** Question: Add another html.Div to show your name,'. The notebook output shows the code executed successfully. The web browser on the right shows the output of the Dash application, which displays 'Hello World' and 'I am Gordon'.
3	Question: Connecting to Data cell has a question - change page size and observe the change in widget controls such as, total number of pages. You will need to update the code, run the cell, capture the screenshot of the output and paste it here. Answer:

```
Explorer (Ctrl+Shift+I) nb
C:\Users\gordo> OneDrive> Desktop> Y3> T2> SIT225 Data Capture> SIT225 2026T2> Web Jupyter Notebook
Generate Code Markdown Colab Run All Run All Restart base (Python 3.11.7)

Connecting to Data

There are many ways to add data to an app: APIs, external databases, local .txt files, JSON
files, and more. In this example, we will highlight one of the most common ways of
incorporating data from a CSV sheet.

# Import packages
# We import the dash_table module to display the data inside a Dash DataTable.
from dash import Dash, html, dash_table

# We also import the pandas library to read the CSV sheet data.
import pandas as pd

# Incorporate data
df = pd.read_csv('gapminder2007.csv')

# Explore data
print(df.head())
print("Data rowsXcols:", df.shape)

# Initialize the app
app = Dash()

# App layout.
# The 2nd item is a table with only 10 rows per page.
app.layout = html.Div([
    html.Div(children='My First App with Data'),
    dash_table.DataTable(data=df.to_dict('records'), page_size=10)
])

# Question: Change page size and observe the change in widget controls

Python
```

My First App with Data

country	pop	continent	lifeExp	gdpPerCap
Afghanistan	31889923	Asia	43.828	974.5803384
Albania	3600523	Europe	76.423	5937.629525999999
Algeria	33333216	Africa	72.301	6223.367465
Angola	12428476	Africa	42.731	4797.231267
Argentina	40301927	Americas	75.32	12779.37964
Australia	20434176	Oceania	81.235	34435.367439999995
Austria	8199783	Europe	79.829	36126.4927
Bahrain	708573	Asia	75.635	29796.04834
Bangladesh	150448339	Asia	64.062	1391.253792
Belgium	10392226	Europe	79.441	33692.60508

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plotly_explore.ipynb
SIT225 2026T2> Web 6> plotly_explore.ipynb> M4 Connecting to Data> Import packages Jupyter Notebook
Generate Code Markdown Colab Run All Run All Restart base (Python 3.11.7)

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# Initialize the app
app = Dash()

# App layout.
# The 2nd item is a table with only 10 rows per page.
app.layout = html.Div([
    html.Div(children='My First App with Data'),
    dash_table.DataTable(data=df.to_dict('records'), page_size=5)
])

# Question: Change page size and observe the change in widget controls
# such as, total number of pages.

# Run the app
if __name__ == '__main__':
    app.run(debug=True, jupyter_mode='tab')

Python
```

My First App with Data

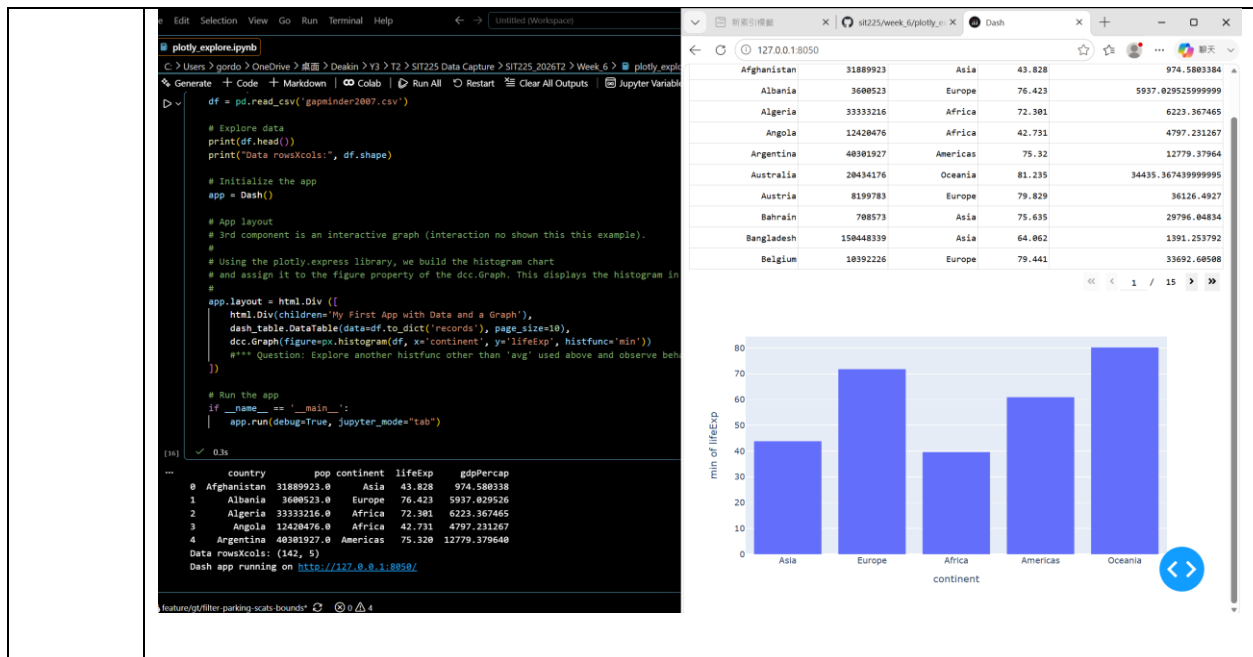
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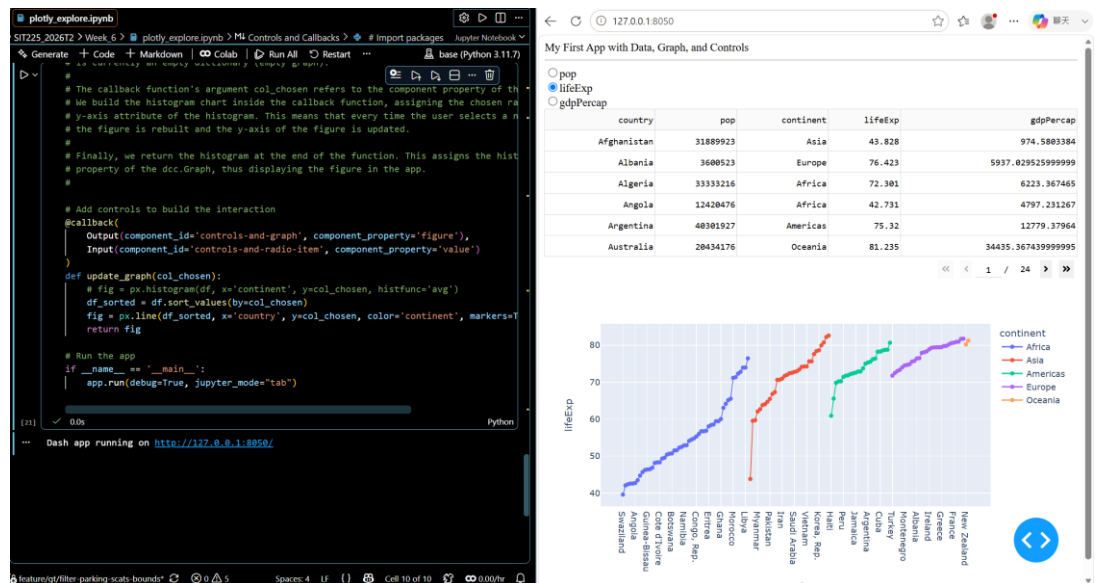
Question: Visualising data cell has a question - explore another histfunc other than 'avg' used above and observe behaviour. You will need to update the code, run the cell, capture the screenshot of the output and paste it here.

Answer:



5 **Question:** Controls and Callbacks cell has a question - use line graphs instead of histogram. You will need to update the code, run the cell, capture the screenshot of the output and paste it here.

Answer:



6 **Question:** Now you have learned how to use Plotly Dash for visualising your data, describe how you will be using this tool for your desired sensor

	monitoring dashboard with a number of sensors including DHT22 or accelerometer data. Answer: plotly dash is a tool to visualize data, since it manage by code base, I can use capture sensors data by Arduino cloud or python script then pass that data to dash so I can see the data in real time. And find out more patten.
7	Question: Convert the Notebook to PDF and merge with this activity sheet PDF. You will need this merged PDF to combine with this week's OnTrack task for submission.